

Machine learning for sample-based quantum diagonalization: generative configuration recovery and the classical-simulability frontier

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Abstract

Sample-based quantum diagonalization (SQD), equivalently quantum-selected configuration interaction (QSCI), has in two years become the pragmatic centre of gravity of pre-fault-tolerant quantum chemistry: a quantum processor samples electronic configurations, and the many-electron Hamiltonian is diagonalized classically in the resulting determinant subspace. Its accuracy is governed entirely by *which* configurations enter that subspace, a selection problem that is a natural target for machine learning and that a coupon-collector bottleneck makes acute. We critically review the resulting ecosystem of generative and learned selectors, from restricted Boltzmann machines and transformer neural quantum states to importance classifiers, learning-to-rank models, and generative circuit design, organizing it by the object each method generates and the importance signal it exploits. The taxonomy exposes one conspicuous gap: a reward-proportional generative-flow-network proposer built for the tail-discovery problem itself. We then confront the question the field turns on, whether the quantum sampler (with or without machine learning) beats classical selected configuration interaction, and report a clean but carefully scoped negative: across the published same-active-space comparisons, strong classical selected CI matches or beats the quantum-sampled subspace, and the flagship shallow circuits now admit polynomial-time classical energy estimation. From this we distil a benchmarking standard, and turn the negative into a regime map. A quantum or generative edge can survive only where the classical importance heuristic that makes an SQD subspace *verifiable* also fails to *construct* it cheaply: plausibly the noise-limited configuration-recovery regime and genuinely multireference chemistry. We frame these as testable predictions and give first controlled, FCI-exact evidence for them: the cheap prior’s rank correlation with the true weights is an order parameter that collapses as multireference character grows, and a two-system noise sweep places the generative crossover only in the multireference regime, not the single-reference one. Finally, we flag learning from quantum experiments, whose classical sample-complexity lower bound is an unconditional theorem, as the one adjacent frontier where a quantum advantage is provable but its bridge to chemistry remains unbuilt.

1. Introduction

The determination of the ground-state energy of a many-electron Hamiltonian is among the oldest and most consequential problems in the physical sciences, underlying our predictive understanding of chemical reactivity, catalysis, and correlated materials. Its difficulty is structural: the dimension of the full configuration-interaction (FCI) space grows combinatorially with the number of electrons and orbitals, and the exact problem is, in the worst case, QMA-complete [1]. The last half-century of quantum chemistry can be read as a sequence of increasingly sophisticated strategies for evading this exponential wall without paying its full cost — coupled-cluster expansions, the density-matrix renormalization group (DMRG), auxiliary-field quantum Monte Carlo (AFQMC), and, most directly relevant here, *selected* configuration interaction (sCI), which builds a compact variational subspace by retaining only the determinants that matter and correcting the remainder perturbatively [2–5].

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The advent of programmable quantum processors reframed the same problem. If the exponential difficulty of electronic structure is a quantum difficulty, perhaps a quantum device can help. The first decade of this hope was dominated by the variational quantum eigensolver (VQE) [6], in which a parameterized circuit is optimized to minimize an energy expectation value. That programme has, for the purpose of demonstrating a scaling advantage, largely run its course: barren plateaus render deep expressive circuits untrainable [7, 8], the measurement cost of estimating chemical-accuracy energies is prohibitive, and — most damningly — the structural conditions that let a variational circuit *avoid* barren plateaus tend also to render it classically simulable [9]. Into the vacuum left by VQE stepped a conceptually different, and strikingly effective, idea.

Sample-based quantum diagonalization. Rather than optimize a circuit to *be* the ground state, one uses a shallow, noise-tolerant circuit only to *sample* the electronic configurations (computational-basis bitstrings, i.e. Slater determinants) that carry appreciable ground-state weight, and then diagonalizes the Hamiltonian classically in the sampled subspace on high-performance hardware. Introduced as quantum-selected configuration interaction (QSCI) by Kanno and co-workers [10] and scaled to quantum-centric supercomputing as SQD by an IBM–RIKEN collaboration [11], the method inherits the variational guarantee of sCI, sidesteps the optimization and measurement pathologies of VQE, and is remarkably robust to hardware noise through a self-consistent configuration-recovery step that repairs symmetry-broken samples. Within two years it has been demonstrated on iron–sulfur clusters at 77 qubits with the entire Fugaku supercomputer in the loop [11, 12], extended to excited states, embeddings, and periodic systems, and pushed, through fragment-based workflows, to a 12,635-atom protein–ligand complex [13]. SQD is, today, the pragmatic centre of gravity of pre-fault-tolerant quantum chemistry.

The machine-learning entry point. The accuracy of the SQD loop is determined entirely by which configurations populate the subspace, and this is where the method is most vulnerable and most interesting. The relevant determinants are heavy-tailed in amplitude; discovering the rare, chemically decisive ones by repeated sampling is a coupon-collector problem, in which the marginal cost of each newly discovered configuration grows sharply [14]. The configuration-recovery step, a probabilistic repair of noise-corrupted samples toward an estimated occupation profile, is a statistical denoising problem. Both are natural targets for machine learning, and the field has responded with striking speed. In the two years since QSCI, generative and learned methods have been proposed for every stage of the loop: restricted Boltzmann machines that learn and resample the distribution of dominant determinants [15, 16]; autoregressive and transformer neural quantum states that propose determinants inside the sample–diagonalize–update loop [17–19]; neural-network classifiers and learning-to-rank models that predict determinant importance [20–22]; and generative circuit design that learns the sampling ansatz itself [23]. This ecosystem has, to our knowledge, never been reviewed, and it is growing faster than any single group can track.

The question the field turns on. Beneath the methodological activity lies a single, contested, and increasingly uncomfortable question: does any of this actually beat classical selected configuration interaction? SQD *is*, after all, a quantum-sampled sCI, and its classical competitors — semistochastic heat-bath CI (SHCI), the perturbatively-selected CIPSI, adaptive sampling CI (ASCI), DMRG, and AFQMC — are mature, fast, and run on a laptop. The critical literature has grown in tandem with the method. Reinholdt and co-workers showed that, even granting SQD a perfect noiseless sampler, its expansions are an order of magnitude less compact than heat-bath CI on the very systems used to showcase it [14]; Gaberle and Jattana found that for strongly correlated lattice models the required configuration count grows exponentially *even with an optimal sampler* [24]; Vaquero-Sabater and co-workers demonstrated that random determinants passed through the classical recovery loop reproduce the accuracy, implying that the classical recovery, not the quantum sampler, is doing the work [25]; and, most strikingly, Belagali and co-workers showed in 2026 that the single-layer local unitary cluster Jastrow circuit SQD samples is classically simulable in polynomial time, reproducing the flagship 77-qubit experiment on a laptop at *lower* energy — at the energy (weak-simulation) level, not by out-sampling the device [26]. The pro-quantum case remains alive but narrow: worst-case instantaneous-quantum-polynomial (IQP) hardness of the sampling circuits [27], and a contested

49-qubit advantage claim on sparse-ground-state Hamiltonians [28].

This review. We take the position that a field cannot mature while its central claim is unsettled, and that the right place to stand is neither uncritical enthusiasm nor dismissal, but a rigorous map of what is known. We make five contributions. First (§2–§3), we give a structured account of the SQD/QSCI method family and the configuration-recovery bottleneck that makes it a machine-learning problem. Second (§4), we provide the first taxonomy of the ML and generative methods proposed for determinant and subspace selection, organized by the object they generate and the importance signal they exploit, and we identify the conspicuous methodological gap: that generative flow networks, purpose-built for diverse reward-proportional sampling over exponential discrete spaces, have not yet been applied to this problem. Third (§5), we confront the quantum-advantage question directly, assembling the classical baselines, the benzene-benchmark accuracy standard, the critique literature, and the 2026 dequantization results into a single defensible verdict. Fourth (§6), we propose a set of benchmarking standards that we contend any future claim in this field must meet. Fifth (§7), we argue that the negative result is a compass, identifying the regimes (noise-limited recovery, genuine multireference character, and the distinct quantum-data-native task of learning from experiments) where generative and quantum methods retain a defensible or, in one case, provable advantage. Throughout, we distinguish sharply between what is proven, what is demonstrated, and what is hoped.

Abbreviations. SQD/QSCI (sample-based quantum diagonalization / quantum-selected configuration interaction); sCI (selected configuration interaction); SHCI (semistochastic heat-bath CI); CIPSI (perturbatively selected CI); ASCI (adaptive sampling CI); FCI (full configuration interaction); DMRG (density-matrix renormalization group); AFQMC (auxiliary-field quantum Monte Carlo); LUCJ (local unitary cluster Jastrow); S-CORE (self-consistent configuration recovery); VQE (variational quantum eigensolver); NQS (neural quantum state); RBM (restricted Boltzmann machine); GQE (generative quantum eigensolver); GFlowNet (generative flow network); EN (Epstein–Nesbet); PT (perturbation theory).

2. Sample-based quantum diagonalization: the method and its family

Scope and method. This review surveys the sample-based quantum diagonalization literature indexed on arXiv (quant-ph, physics.chem-ph) and in the major chemistry and quantum-information journals, together with the community bibliography *Awesome-QSCI*, from the introduction of QSCI in early 2023 to mid-2026. We include the SQD/QSCI method family, the machine-learning and generative methods proposed for determinant or subspace selection, the classical selected-configuration-interaction competitors that constitute the benchmark, and the critique and dequantization literature; we exclude general variational-quantum-eigensolver work except where it bears directly on the sampling step. Because the field is moving on a scale of weeks and much of the primary literature is preprint, we date every claim explicitly and flag its standing — established, demonstrated, or merely hoped.

2.1 The core loop

The object shared by all methods in this review is a hybrid quantum–classical loop with four stages. (i) A parameterized quantum circuit prepares an approximate ground state. In the flagship demonstrations this is the *local unitary cluster Jastrow* (LUCJ) ansatz [29], a hardware-efficient truncation of unitary coupled cluster that alternates orbital rotations with diagonal Coulomb (Jastrow) factors and restricts entangling operations to a physically motivated local connectivity, making it native to heavy-hex superconducting lattices and initializable from classical CCSD amplitudes. (ii) The circuit is measured in the computational basis; under the Jordan–Wigner mapping each length- $2M$ bitstring (for M spatial orbitals) is an electronic configuration, split into an α and a β half. Each measured bitstring is thus a *proposed* Slater determinant, and the collection of distinct bitstrings, with their frequencies, is the raw material the rest of the pipeline processes. (iii) In

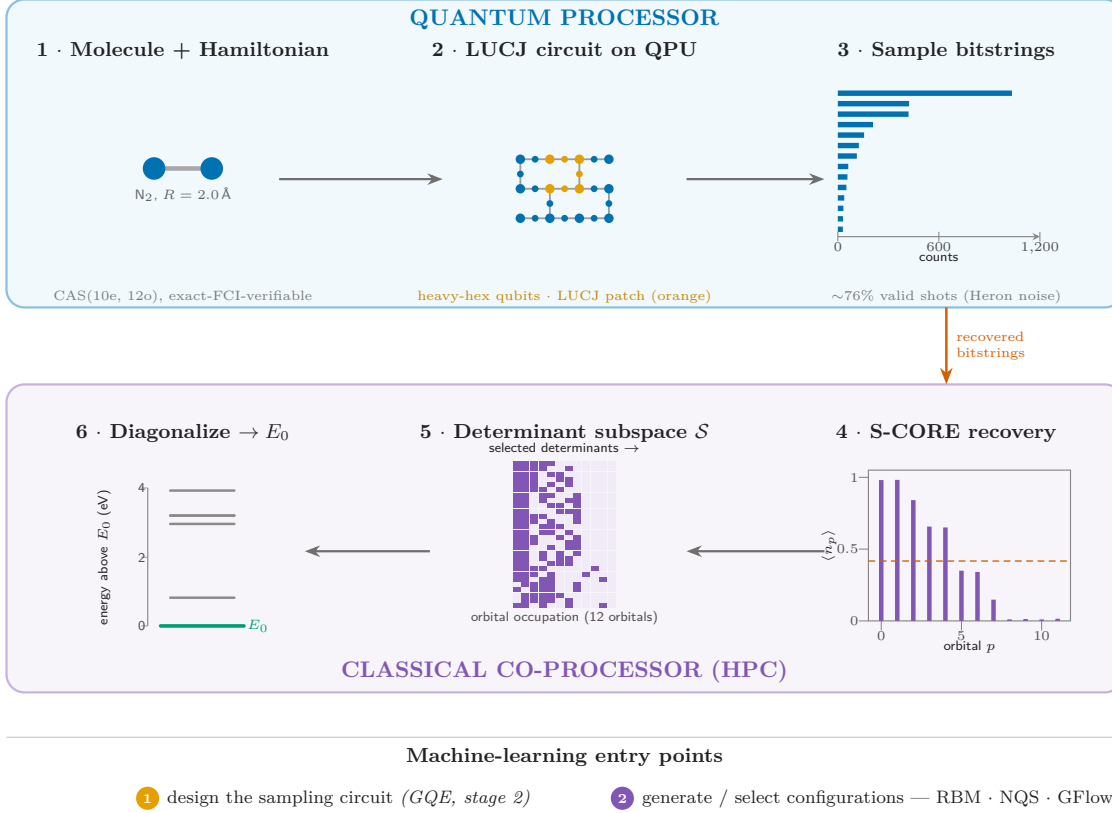


Figure 1. The sample-based quantum diagonalization workflow, with real data at every stage (N_2 , CAS(10e,12o), exact-FCI-verifiable). A shallow LUCJ circuit on the quantum processor (orbital rotations $U(\theta)$ interleaved with diagonal-Coulomb Jastrow gates) is sampled in the computational basis; here 76% of shots survive particle-number post-selection at Heron-calibrated noise. The classical co-processor repairs the broken samples by occupation-weighted S-CORE, assembles the recovered determinants into a subspace $\mathcal{S}$ (each row of panel 5 is one selected configuration’s orbital occupation), and diagonalizes the projected Hamiltonian, whose energy is a variational upper bound converging to the exact FCI value (panel 6). Machine learning augments either the sampler (①, generative circuit design) or the generation/selection of configurations (②). Every panel is real output of the accompanying calculations.

SQD, noise corrupts a large fraction of these bitstrings to the wrong particle number or spin; a *self-consistent configuration recovery* (S-CORE) step repairs them, flipping spin-orbitals toward the mean occupation of the current best eigenvector and iterating to self-consistency [11]; we make the mechanism precise below. (iv) The Hamiltonian is projected onto the recovered determinant subspace and diagonalized by a sparse (Davidson) eigensolver, distributed across high-performance classical hardware; the result is a variational upper bound and a sparse ground-state wavefunction. Formally, the electronic Hamiltonian in second quantization is

$$\hat{H} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} (pq|rs) \hat{a}_p^\dagger \hat{a}_r^\dagger \hat{a}_s \hat{a}_q + E_{\text{nuc}}, \quad (1)$$

whose exact diagonalization in the full configuration-interaction (FCI) basis of M active spatial orbitals with (n_α, n_β) electrons has dimension $\dim \mathcal{H}_{\text{FCI}} = \binom{M}{n_\alpha} \binom{M}{n_\beta}$, growing combinatorially. SQD replaces $\mathcal{H}_{\text{FCI}}$ by a sampled subspace $\mathcal{S} = \text{span}\{|D_x\rangle\}_{x \in \mathcal{S}}$ of Slater determinants and solves the projected eigenproblem for the projector $\hat{P}_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |D_x\rangle \langle D_x|$,

$$(\hat{P}_{\mathcal{S}} \hat{H} \hat{P}_{\mathcal{S}}) \mathbf{c} = E_{\mathcal{S}} \mathbf{c}, \quad E_{\mathcal{S}} \geq E_{\text{FCI}}, \quad (2)$$

the inequality (the Hylleraas–Undheim–MacDonald theorem) being the variational guarantee that makes every SQD energy a rigorous upper bound that improves monotonically as good determinants

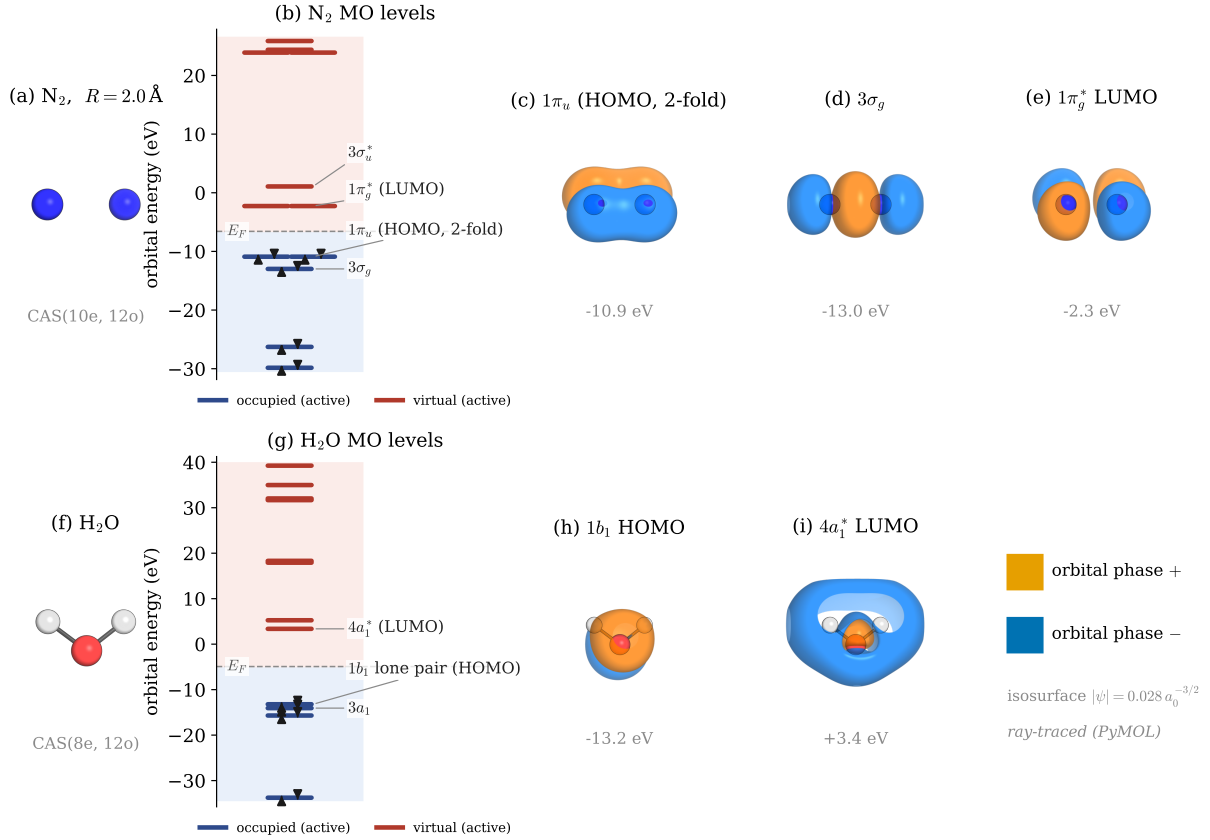


Figure 2. The molecular systems, their active-space orbital-energy structure, and the frontier orbitals. (a,f) Ray-traced ball-and-stick geometries of stretched N_2 ($R = 2.0 \text{ \AA}$, CAS(10e,12o)) and H_2O (CAS(8e,12o)). (b,g) Molecular-orbital energy-level diagrams (restricted Hartree–Fock, eV): occupied active levels (blue) carry their two electrons as up/down arrows, virtual active levels are red, the shaded band is the complete active space, and E_F marks the HOMO–LUMO midpoint; the deep $1s$ core is frozen and off-scale. (c–e) Ray-traced (PyMOL) isosurfaces of the N_2 frontier orbitals — the two-fold-degenerate $1\pi_u$ HOMO (-10.9 eV ; note the axis-containing nodal plane of π symmetry), the non-degenerate $3\sigma_g$ level just below it (-13.0 eV ; cylindrically symmetric about the bond axis), and the $1\pi_g^*$ LUMO (-2.3 eV) — with their RHF energies. The crowding of these frontier levels as the triple bond stretches, and the collapse of the bonding–antibonding gaps ($1\pi_u/1\pi_g^*$, $3\sigma_g/3\sigma_u^*$) toward dissociation, is what makes stretched N_2 strongly multireference: at $R = 2.0 \text{ \AA}$ the FCI natural-orbital occupations of the frontier pair are already $1.32/0.68$ (versus the closed-shell $2/0$) and the Hartree–Fock determinant carries only 34% of the exact weight (§7). (h,i) The H_2O $1b_1$ lone-pair HOMO and $4a_1^*$ LUMO. Isosurfaces at $|\psi| = 0.028 a_0^{-3/2}$; orbital phases orange/blue.

are added. This guarantee is a double-edged asset, and the edge it cuts against is the theme of §5–§7: the same projected-Hamiltonian structure that lets one *certify* an SQD energy from the classical side is what lets one *construct*, screen, and, for the shallow circuits actually run, *simulate* the subspace classically. The variational principle that grounds the method’s appeal is thus also the doorway through which its classical competitors enter. The self-consistent recovery (S-CORE) closes the loop: from the current subspace eigenvector it forms the mean spin-orbital occupations $\langle n_p \rangle = \sum_{x \in \mathcal{S}} |c_x|^2 [x]_p$ and repairs each symmetry-broken bitstring by flipping spin-orbitals with probability increasing in the deviation $|[x]_p - \langle n_p \rangle|$, iterating recovery $\rightarrow$ diagonalization $\rightarrow$ occupation-update to self-consistency. Because the whole quantitative analysis of this review lives in an active space small enough for exact FCI (N_2 and H_2O in CAS(10e,12o)/(8e,12o), whose active-space orbitals are shown in Fig. 2), every energy we quote below is an *error against exact truth* within that model space, not against another approximation.

Two lineages coined the idea nearly simultaneously and use different names for it. *Quantum-selected*

configuration interaction (QSCI) was introduced by Kanno and co-workers [10], now published in *Physical Review Research*, and framed the method as “the quantum computer selects the subspace, the classical computer diagonalizes.” *Sample-based quantum diagonalization* (SQD) is the IBM–RIKEN realization at quantum-centric-supercomputing scale [11], adding the self-consistent recovery loop and the tight coupling to an HPC back end. We use SQD and QSCI interchangeably, noting the recovery step as the principal methodological difference and the natural home of the machine-learning contributions reviewed in §4. Crucially, because the final step is an exact diagonalization in a fixed subspace, the energy is variational: it can only be an upper bound to the true ground state, and it improves monotonically as good determinants are added — a property that both grounds the method’s appeal and, as §5 will show, sets the exact bar its classical competitors must be measured against.

2.2 A taxonomy of variants

In two years the SQD/QSCI family has branched into some fifty distinct variants; we organize them by the stage of the loop they modify. *Input-state variants* replace the LUCJ sampler: ADAPT-QSCI grows the state one Pauli generator at a time without a variational optimization [30]; time-evolved QSCI (TE-QSCI) and Hamiltonian-simulation QSCI (HSB-QSCI) generate multi-order excitations from real-time evolution rather than circuit optimization [31, 32]; and generative circuit design learns the sampler itself (§4.5). *Krylov variants* (sample-based Krylov quantum diagonalization, SKQD, and its randomized provable-convergence descendant SqDRIFT) build a Krylov subspace from sampled time-evolved states, importing phase-estimation-like convergence guarantees at low depth [33, 34]. *Recovery and selection variants* attack the coupon-collector bottleneck directly: amplitude amplification of unseen bitstrings [35], active-sampling by an Epstein–Nesbet acquisition function [36], and (the focus of §4) machine-learned recovery. *Qubit-reduced variants* halve the register through half-qubit encodings, seniority-zero sampling, entanglement forging, or neural fermionic compression [37–40], and a CI-matrix reformulation (CIM-QSCI) pushes further to a logarithmic $\lceil \log_2 N \rceil$ qubit count [41]. *Embedding variants* reach extended systems by using SQD as the impurity or fragment solver inside density-matrix embedding, localized-active-space SCF, or DFT embedding [13, 42]. *Hybrid post-processing variants* recover dynamical correlation missed by the finite subspace, using the SQD wavefunction as an AFQMC trial, a tailored-CC reference, or a multireference-PT2 anchor [43, 44]. A full annotated tabulation, with citations, systems, and largest scale, is given in the accompanying dataset.

2.3 Hardware demonstrations and honest scale

The method’s hardware trajectory is genuinely impressive and worth stating precisely, because it is often conflated with an advantage claim it does not make. The founding demonstration reached N_2 at 58 qubits and the iron–sulfur clusters [2Fe-2S] and [4Fe-4S] on 45- and 77-qubit registers, with classical diagonalization distributed across up to 6,400 nodes of Fugaku [11]; the [4Fe-4S] run, a (54e,36o) active space (72 Jordan–Wigner spin-orbitals, the balance of the register serving the heavy-hex LUCJ mapping), was the largest quantum-computed at the time. A full-scale closed-loop follow-up used the entire 152,064-node Fugaku machine with an on-premises Heron processor and improved the [4Fe-4S] energy by 278 mHa over the 2024 result [12]. Fragment-based embeddings then pushed the workflow, in 2025–2026, from a ~ 300 -atom miniprotein to a 12,635-atom protein–ligand complex [13]. These are landmark demonstrations of *scale beyond exact diagonalization*. They are not, and are not claimed by their authors to be, demonstrations of an advantage over the best classical methods, a distinction §5 makes central.

Table 1. The SQD/QSCI method family (curated; the full annotated list is maintained in the state-of-the-art dataset). “Largest scale” states hardware qubit counts or active spaces as reported.

Category	Method	Key idea	Ref.	Largest scale
Core	QSCI	Quantum samples the subspace; classical diagonalization	[10]	8 qubits
Sampler/input	SQD + S-CORE	Self-consistent configuration recovery; quantum-centric supercomputing	[11]	77 q / (54e,36o)
	ADAPT-QSCI	Iterative input-state growth, no VQE	[30]	small molecules
	TE-/HSB-QSCI	Time-evolved / Hamiltonian-simulation sampling state	[31,32]	36 q (carbyne)
	Generative-QSCI	Transformer (GQE) designs the sampling circuit	[23]	32 q (N ₂)
Krylov	SKQD / SqDRIFT	Sampled Krylov subspace; provable convergence	[33,34]	49 q
Recovery/select	SQD-AA	Amplitude-amplify unseen bitstrings (quadratic gain)	[35]	model systems
	ML recovery	Learned generation/classification (see Table 2)	—	see Table 2
Excited/orbital	ext-/oo-SQD	Extra projection for excited states; orbital optimization	[45]	CH ₂ 52 q
Symmetry/compact	Compact-QSCI	Space-group symmetry; time-evolved compact subspaces	[46,47]	SiH ₄ 42 q
Qubit-reduced	HCI-HSQD	Half-qubit; seniority-zero; neural compression	[37]	Fe-S (54e,36o)
Embedding	DMET-/LAS-SQD	SQD as fragment / impurity solver	[13]	protein 12,635 atoms
Hybrid post-proc.	AFQMC-SQD; PT2	Recover dynamical correlation on the SQD reference	[43,44]	N ₂ ; [2Fe-2S]

2.4 Software and ecosystem

The method is unusually well tooled for its age, which partly explains its rapid uptake. IBM maintains `qiskit-addon-sqd` (with an HPC C++ variant) and `ffsim` for fermionic-circuit simulation and LUCJ construction; QunaSys maintains `quri-parts-qsci` within the QURI SDK and, notably, a continuously updated community bibliography, *Awesome-QSCI*, that functions as a living review of a field moving too fast for the conventional publication cycle. NVIDIA’s CUDA-Q documents SKQD as a reference application and provides GPU primitives for the classical diagonalization back end. This tooling maturity is a double-edged asset: it lowers the barrier to method development — including the machine-learning methods of §4 — but it also lowers the barrier to producing demonstrations that are not benchmarked against the strong classical baselines of §5.

3. The configuration-recovery bottleneck: the machine-learning entry point

3.1 A coupon-collector law

The efficiency of the entire loop reduces to a single question: how many distinct, *important* determinants can be placed in the subspace, and at what cost? The answer is governed by the amplitude distribution of the exact ground state, which is heavy-tailed. In stretched N₂ in a (10e,12o) active space, for instance, 90% of the exact FCI weight is carried by roughly 1.4% of the single-spin strings (about a dozen of the 792), with per-configuration weights spanning many orders of magnitude. Sampling from such a distribution to *discover* the low-amplitude but chemically decisive

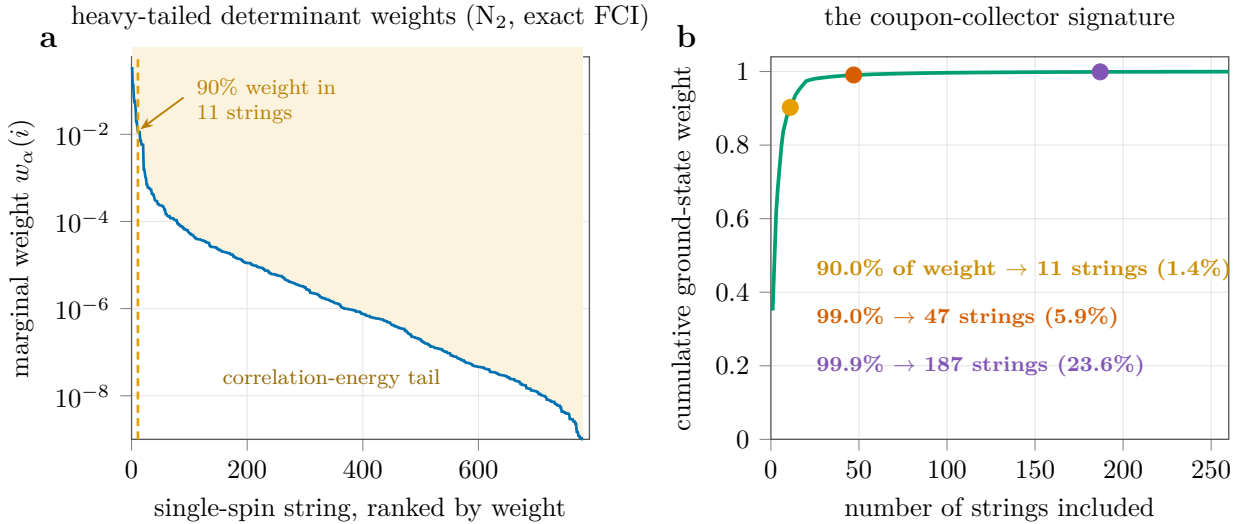


Figure 3. The coupon-collector bottleneck, computed exactly. N₂ CAS(10e,12o), from the exact FCI wavefunction. *Left:* the marginal single-spin-string weight $w_\alpha(i) = \sum_\beta |c_{i\beta}|^2$ falls across roughly eight orders of magnitude, from 0.35 down to $\sim 10^{-9}$. *Right:* 90% of the ground-state weight is carried by about a dozen of the 792 strings (roughly 1.4%), but the correlation-energy tail is spread over hundreds of rare strings — the configurations a sampler must pay to discover.

determinants is a coupon-collector problem with non-uniform probabilities: as sampling proceeds, an overwhelming fraction of draws re-select already-seen configurations, and the number of newly discovered determinants per shot collapses. Reinholdt and co-workers made this quantitative and damaging: for N₂ the discovery rate falls from ~ 0.1 determinants per shot at 10^3 shots to $\sim 5 \times 10^{-4}$ at 10^9 , and reaching micro-hartree precision extrapolates to on the order of 10^{14} samples [14]. This is the bottleneck that machine learning is invoked to break: if a learned model can *generate* the important tail rather than wait to sample it, the coupon-collector wall might be circumvented. Whether it can — and whether doing so beats simply running a classical selected-CI heuristic — is the question of §4 and §5.

The effect is exact and computable. Writing the marginal weight of a single-spin string i as $w_\alpha(i) = \sum_\beta |c_{i\beta}|^2$ and its sampling probability as p_i , the expected number of shots required to first observe every string in a target set $\mathcal{T}$ is bounded above by the coupon-collector sum (the leading inclusion–exclusion term, tight in the heavy-tail regime the argument invokes)

$$\mathbb{E}[\text{shots to collect } \mathcal{T}] \leq \sum_{i \in \mathcal{T}} \frac{1}{p_i} \xrightarrow{p_i \rightarrow 0} \infty, \quad (3)$$

which upper-bounds the exact expectation (the leading term of an inclusion–exclusion series) and diverges as the heavy tail is approached. Fig. 3 computes this directly from the exact FCI wavefunction of N₂ CAS(10e,12o): the same handful of dominant strings carry almost all the weight, but the correlation-energy tail that decides chemical accuracy is spread over hundreds of rare strings — precisely the configurations whose p_i make the sum above large, and precisely where a learned generator is supposed to earn its keep.

3.2 Recovery as denoising, and the peculiar role of noise

On real hardware the bottleneck is compounded by noise, and here the picture is subtle in a way that is central to the field’s clear-eyed self-assessment. Under readout and gate noise, the majority of measured bitstrings violate particle-number conservation and would be discarded by naive post-selection: for N₂ in a large active space on hardware, only about 0.17% of raw samples survive [25]. This survival fraction falls steeply with register width, since each additional qubit is

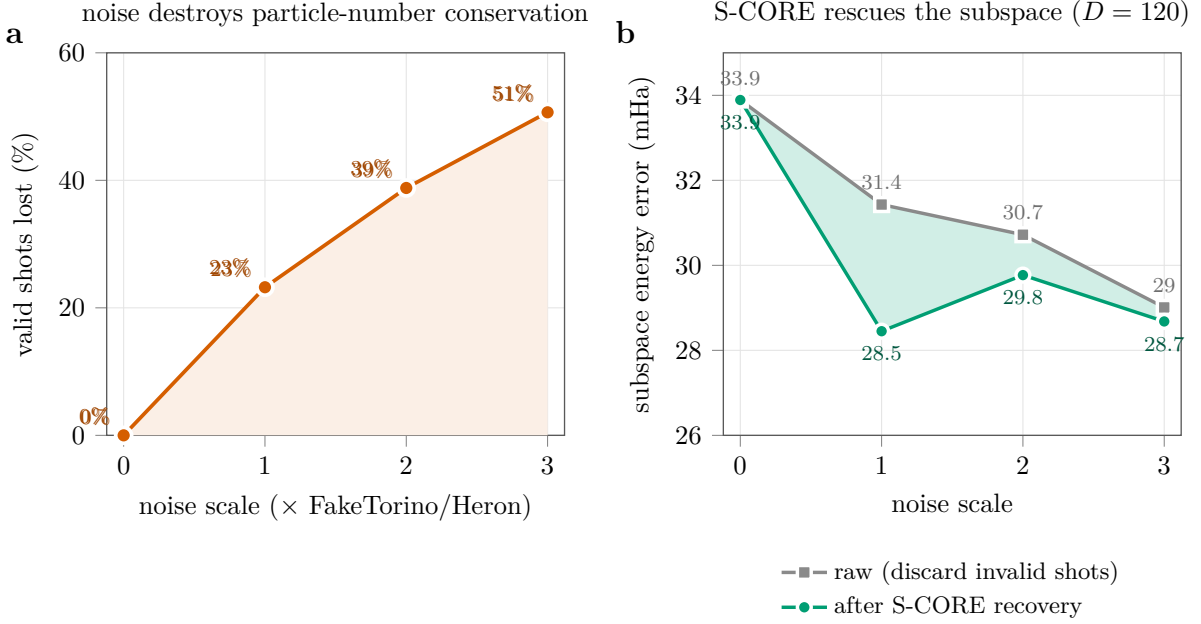


Figure 4. Configuration recovery under backend-calibrated noise (FakeTorino / Heron r1 rates; asymmetric readout + depolarizing). *Left*: the fraction of particle-number-valid shots collapses as the noise scale grows. *Right*: at fixed dimension $D = 120$, S-CORE recovery (green) lowers the subspace energy error relative to discarding the invalid shots (grey) — most at moderate noise (~ 3 mHa at $1\times$, shrinking to ~ 0.3 mHa at $3\times$) — while the raw curve (grey) drifts steadily downward and the recovered curve (green) drops sharply then stays low as noise broadens Hilbert-space exploration, both consistent with the “noise can help” effect. Note the recovered subspace stays well above chemical accuracy here (~ 28 – 34 mHa at this fixed, illustrative D): recovery salvages otherwise-discarded samples, it does not by itself reach chemical precision.

another channel for a particle-number-violating readout error: our illustrative 12-orbital N_2 CAS retains $\sim 76\%$ of shots at $1\times$ Heron noise (Fig. 1), whereas a large hardware active space over many more qubits retains a fraction of a percent, the same mechanism two orders of magnitude apart in scale. The self-consistent configuration recovery step is precisely a statistical imputation model that rescues these corrupted samples, and it is the true workhorse of SQD’s noise tolerance. Strikingly, Vaquero-Sabater and co-workers showed that noise can *help*: by scattering samples beyond the biased support of the ideal ansatz, moderate noise broadens Hilbert-space exploration, and recovery starting from *uniformly random* bitstrings reaches accurate energies within a couple of iterations. The implication, that the classical recovery loop (not the quantum sampler) is responsible for the accuracy, is one of the sharpest results in the literature and is examined in §5. For the present section the point is narrower: the recovery step is a learned/statistical model, and it is the natural site for the generative methods of §4, whose promise is to replace an occupancy-weighted heuristic repair with a model that has learned the structure of the important subspace. Fig. 4 makes the mechanism concrete on backend-calibrated noise: as the device-noise scale grows the fraction of particle-number-valid shots collapses, yet S-CORE recovery still lowers the fixed-dimension subspace error by imputing the broken samples back onto valid configurations, with its largest benefit at moderate noise.

3.3 Why the standard error-mitigation toolkit does not transfer

A methodological subtlety that the field has had to internalize, and that a review should make explicit, is that SQD is a *per-sample* post-selection pipeline, not an expectation-value pipeline. This distinction determines which of the near-term error-mitigation techniques transfer. Post-selection on particle number and spin, self-consistent recovery, and symmetry projection via classical shadows [48] operate on individual bitstrings and are central to the method. In contrast, the techniques that dominate IBM’s utility-scale expectation-value experiments (twirled readout error extinction, TREX; zero-

noise extrapolation, ZNE; and probabilistic error cancellation, PEC) mitigate errors *in expectation values*, not in individual samples, and therefore do not transfer to a pipeline that needs valid configurations rather than corrected averages. Dynamical decoupling and Pauli twirling give only marginal, inconsistent benefit once the recovery loop has converged [49]. The practical consequence, relevant to §6, is that a claim about SQD’s hardware performance must be benchmarked with the mitigation stack it actually uses (recovery + symmetry post-selection), not the expectation-value stack, and that the shot budget saturates surprisingly early, optimal at 10^3 – 10^4 shots, with 10^5 performing *worse* [49]. This early hardware saturation does not contradict the $\sim 10^{14}$ -sample extrapolation of §3.1: that figure is the noiseless discovery limit for the exact heavy tail, whereas this one is the point at which, on noisy hardware with recovery already converged, additional shots mostly re-collect configurations the recovery step has restored rather than discovering new tail determinants.

4. Machine learning and generative methods for determinant and subspace selection

The proposition uniting this section is simple to state: if the important determinants are heavy-tailed and expensive to discover by sampling (§3.1), replace or augment the raw sampler with a *model* that has learned where they are. The proposition is also older than SQD: Coe’s machine-learning configuration interaction (MLCI) trained a neural network on-the-fly to predict which unseen determinants a classical selected-CI loop should add, on CO and stretched water, as early as 2018 [50, 51]. What SQD added is a reason to revisit the idea urgently and at scale, and a quantum sampler to compete with or complement. The result is a fast-moving ecosystem that, to our knowledge, has not been surveyed. We organize it along two axes, the *object* a method generates (a determinant, a subspace, a circuit) and the *importance signal* it is trained on (CI-coefficient magnitude, a perturbative estimate, quantum-sample frequency, or the diagonalized energy itself), and we treat the four families in turn, closing with the conspicuous gap the taxonomy exposes.

4.1 A taxonomy by generated object and importance signal

Table 2 and Fig. 5 arrange the methods on the two axes. Two observations organize the rest of the section. First, the field is converging, from several directions, on a single algorithmic skeleton: *propose configurations, diagonalize in their span, use the result to improve the proposer, iterate*. This is not a coincidence but an attractor: the variational guarantee (§2.1) makes any proposal cheap to *score* by exact diagonalization, and a cheap exact score is precisely what turns configuration selection into a learning problem, pulling classical and quantum approaches alike toward the same loop. The consequence reframes the field: the SQD loop is *proposer-agnostic*, the quantum sampler one proposer among many (heat-bath screening, an RBM, an autoregressive network, a diffusion model, a GFlowNet), and the scientific question reduces to *which proposer wins in which regime*. Read this way, the GFlowNet whitespace of §4.6 is not one more idea but the one systematic gap in the space of proposers the attractor admits. Second, almost every method to date is a *distribution-matching* or *classification* model (it learns to reproduce, or to label as important, the determinants it has already seen); none, before the gap discussed in §4.6, is a *reward-proportional amortized sampler* explicitly built to place mass on diverse, high-importance determinants it has *not* yet seen, which is precisely the tail-discovery problem the coupon-collector law makes hard.

4.2 Restricted Boltzmann machines for generative recovery

The most direct instantiation of “learn the distribution of important determinants and resample it” uses a restricted Boltzmann machine (RBM). The Maitra group at IIT Bombay is the most active locus here. Their PIGen-SQD couples a physics-informed generative model to the SQD recovery step, using implicit low-rank tensor decompositions to steer the generator into the dominant sector,

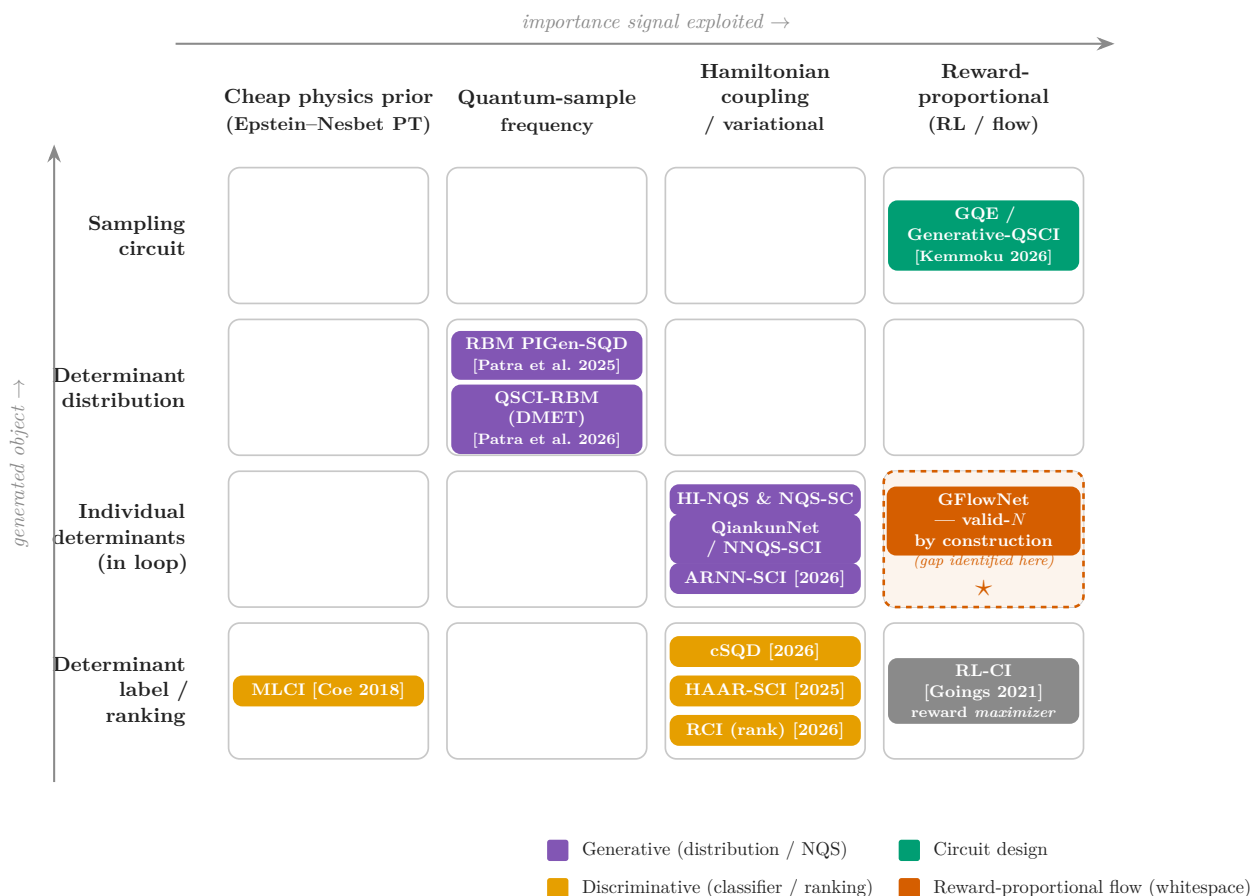


Figure 5. A taxonomy of machine-learning methods for determinant and subspace selection in SQD, organized by the object each method generates (rows) and the importance signal it exploits (columns). Generative models (purple) learn a distribution over determinants; discriminative models (orange) classify or rank them; circuit-design methods (green) learn the sampler itself. The reward-driven column (right) is occupied only by the reinforcement-learning *maximizer* RL-CI [Goings 2021] (grey); the reward-*proportional* flow cell (dashed) that diverse tail discovery calls for is the methodological whitespace this review identifies — a generative-flow network that constructs valid-*N* determinants in the loop, unoccupied as of mid-2026.

and, notably among the ML-for-SQD works, is validated on real IBM Heron hardware [15] (the transformer-QSCI line of [55] is the other hardware-executed example, on Zuchongzhi 3.1). Its density-matrix-embedding successor, QSCI-RBM, learns the distribution of dominant determinants from quantum samples and generates high-probability configurations, reaching chemical accuracy on a SARS-CoV-2 main-protease-inhibitor complex while accessing only $\sim 4\%$ of the configuration subspace, against $\sim 20\%$ for standard SQD [16]. An independent lineage, building on generative-ML-in-configuration-space [56], uses an interpretable RBM to solve the CI problem directly, recovering 99.99% of the correlation energy with orders of magnitude fewer determinants [57]. The strength of the RBM route is that it is a genuine generative model of the determinant distribution; its limitations, relevant to §4.6, are that Gibbs sampling is iterative and mode-seeking, and that particle-number and spin symmetry are enforced by post-hoc filtering rather than by construction. (We note, for completeness and because the review record should be accurate, that an early QSCI-RBM preprint was withdrawn by arXiv administrators for a licensing-rights reason (not a scientific retraction) and its content re-appeared, expanded, in the density-matrix-embedding paper above.)

4.3 Autoregressive and transformer neural quantum states in the loop

A second, and currently the most vigorous, family replaces the quantum sampler with an autoregressive or transformer *neural quantum state* (NQS) [58] that generates occupation-number

Table 2. Machine-learning and generative methods for determinant/subspace selection, organized by the object generated and the importance signal exploited. HW? indicates whether a quantum-hardware demonstration was reported.

Method	Generated object	Importance signal	HW?	Ref.
MLCI	determinant (add/skip)	on-the-fly NN prediction	classical	[50]
PIGen-SQD	determinants (recovery)	quantum-sample freq. + physics prior	IBM Heron	[15]
QSCI-RBM (DMET)	determinant distribution	learned Born distribution (RBM)	classical	[16]
HI-NQS	determinants (in loop)	PT score + distilled eigenvector	classical GPU	[17]
NQS-SC	selected-config energy	variational, on selected set	classical	[18]
NNQS-SCI / QiankunNet	determinants	autoregressive Born + de-dup	classical HPC	[19, 52]
ARNN-SCI	determinants (in loop)	autoregressive NQS sampling	classical	[53]
cSQD	determinant label	binary classifier + active learning	classical (+SQD)	[20]
HAAR-SCI	determinants	Hamiltonian coupling (gated transformer)	classical GPU	[21]
RCI (learning-to-rank)	determinant ranking	pairwise rank (transformer)	classical	[22]
Generative-QSCI (GQE)	circuit (not determinants)	QSCI subspace energy (RL policy)	classical sim.	[23]
RL-CI	determinant (add/skip)	RL policy, reward- <i>maximizing</i>	classical	[54]
GFlowNet (proposed)	determinant, valid-N by construction	reward-<i>proportional</i>: fused EN + sample freq.	— (whitespace)	<i>this work, §4.6</i>

bitstrings by construction with fixed particle number and exact ancestral sampling. The striking development of 2025–2026 is that this classical community has converged, independently, onto the SQD paradigm. HI-NQS embeds a dual-channel transformer (with explicit spin-up/spin-down cross-attention encoding fermionic structure) inside a sample–diagonalize–distil loop that its authors explicitly call sample-based quantum diagonalization, driven by a neural network rather than a circuit, and reports roughly half the determinant count of CIPSI on an N_2 active-space series [17]; a parallel autoregressive-NQS selected-CI, ARNN-SCI, uses the network to guide subspace expansion directly [53]. Solanki, Ding, and Reiher’s NQS-SC evaluates the energy directly on a selected-configuration set and argues, on grounds of systematic improvability, that it should replace variational NQS as the default for electronic structure [18]. At the high-performance-computing end, the QiankunNet line scales a transformer NQS-driven selected-CI engine to Hilbert spaces beyond 10^{14} , with GPU-accelerated de-duplication and coupled-configuration kernels [19, 52]; a transformer-refined QSCI in this line has now been executed on the Zuchongzhi 3.1 superconducting processor, reaching chemical accuracy on a nitrogenase P-cluster model [55]. These methods are, in effect, a classical mirror of SQD: the same loop, the same subspace diagonalization, but with a trained network as the proposer. Their relevance to a review of *quantum*-sampled diagonalization is twofold: they are the most natural classical baseline against which any quantum-sampling advantage must be measured (§5), and their architectural choices (symmetry-by-construction, distillation of the diagonalized eigenvector back into the proposer) are the design vocabulary any quantum-integrated learned proposer will draw on.

4.4 Neural-network classifiers and learning-to-rank selectors

A third family declines to model the full distribution and instead learns only to *rank* or *classify* determinants by importance — a lighter, and in practice highly effective, target. Zeni and co-workers recast determinant selection as binary classification inside an active-learning loop, with a classical variant (cHCI) cutting per-iteration memory roughly fivefold and a quantum–classical variant (cSQD) converging in markedly fewer SQD iterations [20]. HAAR-SCI samples determinants autoregressively with a gated transformer and retains only those with the largest Hamiltonian couplings via GPU min-heap kernels, reaching a mean absolute error of 0.51 mHa across eighteen molecules and, notably, beating semistochastic heat-bath CI on an iron–sulfur cluster while retaining under 0.01% of the Hilbert space [21]. A learning-to-rank formulation with a transformer reaches chemical accuracy on iron–sulfur clusters using 12% of the CI space and reports beating SHCI on the notoriously multireference Cr₂ [22]; and a natural-orbital-based neural-network CI selector chooses determinants using approximate natural orbitals drawn from intermediate many-body eigenstates rather than Hartree–Fock orbitals [59], building on the same group’s earlier neural-network selective CI, which recovered the N₂ correlation energy from $\sim 4 \times 10^5$ rather than $\sim 10^{10}$ determinants [60]. These results matter for the honest assessment of §5 in a specific way: several of them beat *classical* selected-CI — but they do so with a *classical* learned model, using no quantum hardware, which sharpens rather than resolves the question of what the quantum sampler adds.

4.5 Generative circuit design

A fourth family is generative in a different sense: rather than generate determinants, it generates the *circuit* that produces them. The generative quantum eigensolver (GQE) trains a transformer policy (a GPT-style model over circuit tokens) to design ansätze with desired properties, sidestepping the barren-plateau pathology of gradient-based variational optimization [61]. Applied to QSCI, this becomes generative circuit design in which the transformer is trained on the QSCI subspace energy to produce the state-preparation circuit whose samples define the subspace; on N₂ up to 32 qubits it reaches chemical accuracy with 98% fewer two-qubit gates than first-order Trotterization and yields subspaces roughly half the size of heat-bath CI in strongly correlated regimes [23]. This is the closest existing work to “a generative model that produces the SQD sampler,” and, a point we return to in §4.6, it deliberately uses a transformer policy trained by reinforcement-style objectives, not a reward-proportional generative sampler.

4.6 The generative-flow-network gap, and the transferable toolbox

Surveying the four families above against the machine-learning literature exposes a conspicuous and, we argue, consequential gap. Generative flow networks (GFlowNets) [62, 63] are amortized samplers purpose-built to generate compositional discrete objects with terminal probability proportional to a reward, favouring *diverse* high-reward states rather than collapsing onto the mode, a property established both empirically and through their equivalence (in expected gradient) to a hierarchical variational inference that avoids the mode-seeking of reverse-KL objectives [64]. They have been applied across molecular graphs, crystals, biological sequences, combinatorial optimization, and Bayesian structure learning, and in the past year to adjacent quantum tasks — though here the generative-sampler families must be kept distinct: a conditional *normalizing* flow (a different generative paradigm, not a GFlowNet) warm-starts the variational quantum eigensolver in Flow-VQE [65], whereas GFlowNets *proper* have been applied to Pauli-measurement grouping [66]. Yet an exhaustive search across every relevant term (GFlowNet with configuration interaction, determinant, selected-CI, Slater determinant, SQD, QSCI, Fock space) returns, as of mid-2026, *no* application of a GFlowNet to determinant or subspace selection for configuration interaction. The one reward-driven CI selector to date is the reinforcement-learning configuration interaction of [54], which trains a policy to *maximize* the value of the determinants it adds; this occupies the reinforcement-learning, mode-seeking half of the reward-driven column and sharpens rather than fills the gap, because

tail discovery calls for a reward-*proportional*, diversity-preserving sampler, not a maximizer. The nearest generative work is the generative-circuit QSCI of §4.5, which uses a transformer policy and deliberately not a GFlowNet. The gap is not an accident of terminology but a genuine methodological opening: a determinant is a fixed-particle-number occupation bitstring built one orbital at a time (precisely the compositional discrete object a GFlowNet constructs), and the correlation energy lives in the *many small-weight determinants of the tail*, the regime where a reward-proportional sampler that resists mode collapse is exactly the right tool and a reward-maximizing search is exactly the wrong one. We are careful about the scope of the resulting novelty claim. That a *generative* model helps SQD is not novel (§4.2–§4.5); that noise can help recovery is not novel (§3.2). What is unclaimed is the specific combination (a GFlowNet whose forward construction of determinants is masked to the fixed-particle-number sector by construction, trained to sample proportional to a fused importance reward) together with the controlled, exact-ground-truth study (§6) that would establish *when* such a proposer helps. The adjacent literature already supplies the toolbox for it: energy-based GFlowNets as amortized samplers of unnormalized discrete distributions [67], “inexpensive-reward” pretraining that mirrors the cheap-perturbative-estimate-then-expensive-diagonalization structure of the problem [68], temperature-conditional reward tempering that is a learnable analogue of CIPSI’s threshold schedule [69], and discrete-diffusion set-selection models for which a determinant subspace is literally a binary selection mask [70]. Whether any of these transfers *usefully*, and whether it survives the classical baselines of §5, is, we argue, the central open experimental question of the subfield.

Concretely, such a proposer needs a reward that is cheap and FCI-free. The natural choice — the same criterion CIPSI and heat-bath CI use to grow their subspaces — is the Epstein–Nesbet first-order coefficient of each determinant relative to the Hartree–Fock reference,

$$c_i^{(1)} = \frac{\langle D_i | \hat{H} | D_{\text{HF}} \rangle}{E_{\text{HF}} - \langle D_i | \hat{H} | D_i \rangle}, \quad R(i) \propto (c_i^{(1)})^2, \quad (4)$$

computable with a single $\hat{H}|D_{\text{HF}}\rangle$ matrix–vector product and the Hamiltonian diagonal. By the Slater–Condon rules $\langle D_i | \hat{H} | D_{\text{HF}} \rangle$ is nonzero only for determinants differing from the reference by at most two spin-orbitals, so this single-reference prior is structurally blind to the triple and higher excitations that carry weight in a strongly multireference wavefunction — the physical origin of the imperfect ($\rho \approx 0.64$, §7) rank correlation, and a ceiling that no amount of proportional sampling of the prior can lift. A GFlowNet with forward policy $P_F(a_t | s_{t-1}; \theta)$ builds a determinant as a trajectory $\tau : s_0 \rightarrow s_1 \rightarrow \dots \rightarrow x$ that places electrons one orbital at a time — masked so that x is always a valid n_α -electron string — and is trained by the trajectory-balance objective

$$\mathcal{L}_{\text{TB}}(\tau) = \left(\log Z_\theta + \sum_t \log P_F(a_t | s_{t-1}; \theta) - \log R(x) \right)^2, \quad (5)$$

whose global minimizer samples terminal states in exact proportion to reward, $P_\theta^\top(x) \propto R(x)$, the diverse, tail-covering behaviour the coupon-collector regime demands. Fig. 6 evaluates exactly this construction on the FCI-verifiable N_2 system, over five seeds. Given only the cheap, FCI-free reward, the GFlowNet’s diversity lets it beat blind uniform sampling by a wide margin; but, averaged over seeds, it does *not* track the exact oracle, and, tellingly, it does not even beat the *deterministic* greedy top- K selection by the *same* cheap reward (192 ± 19 vs 41.3 mHa at $D=120$). Sampling a cheap reward proportionally, however diversely, cannot outperform simply taking its top determinants when the reward correlates only ≈ 0.64 (Spearman) with the truth: the proposer inherits the ceiling of the signal it is trained on. (An earlier single-seed run of this figure suggested a near-oracle GFlowNet; the multi-seed standard of §6(v), applied to our own figure, corrected it.) Naive i.i.d. importance sampling from the peaked reward, by contrast, collapses onto a handful of modes and saturates below dimension 30, the coupon-collector wall in miniature. The lesson is not that the GFlowNet is worthless but that a learned proposer trained on a cheap classical reward can only reach that reward’s ceiling; the case for it therefore rests on the two regimes where that ceiling lifts (the fused

A cheap, FCI-free GFlowNet beats blind sampling but not the classical selector
(N_2 ; 5-seed mean $\pm$ s.d., greedy top- K deterministic)

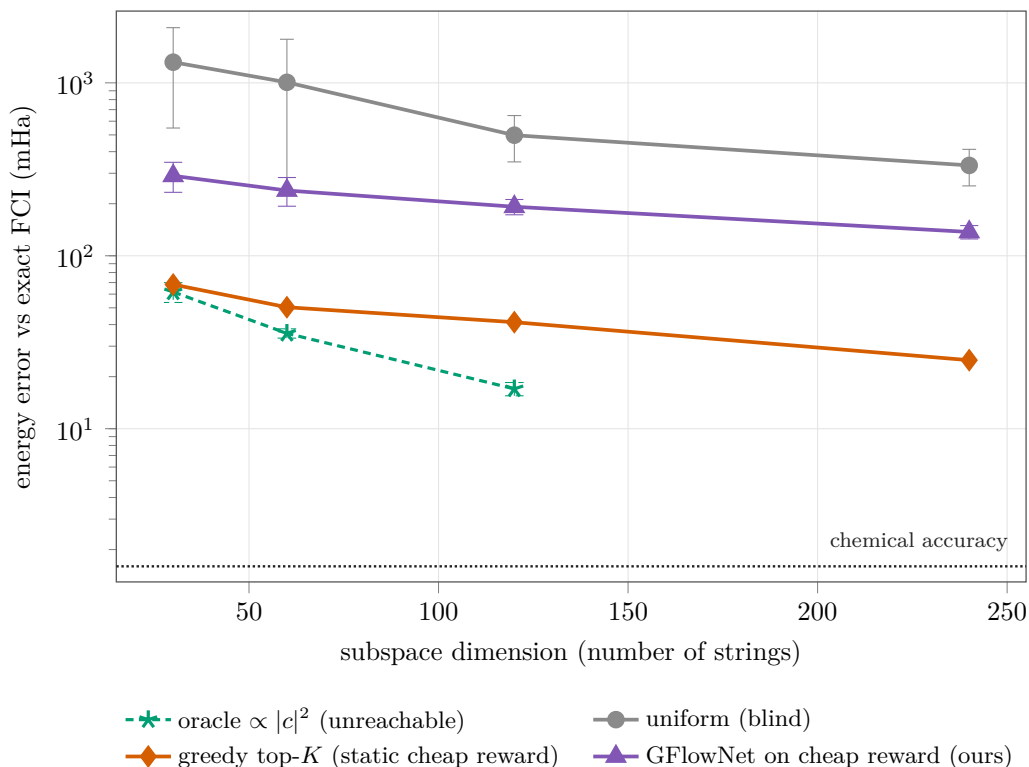


Figure 6. Compactness under a cheap, FCI-free reward (N_2 , exact FCI; 5-seed mean $\pm$ s.d.). Energy error vs subspace dimension for blind uniform sampling, deterministic top- K selection by the static cheap Epstein–Nesbet reward, the GFlowNet trained on the same (tempered, $\beta=0.5$) cheap reward, and the exact $\propto |c|^2$ oracle. Averaged over five seeds, the ordering is unambiguous, and it is the ordering the paper’s verdict predicts: the *deterministic classical* greedy selector (41.3 mHa at $D=120$) beats the GFlowNet (192 ± 19 mHa), which in turn beats only blind uniform sampling (498 ± 149 mHa); the exact oracle (17.0 ± 1.5 mHa) is out of reach of all three. The GFlowNet’s diversity buys a large margin over blind sampling but none over the cheap classical selector it shares a reward with, consistent with the Spearman rank correlation of only ≈ 0.64 between that cheap reward and the true weights (§7). An earlier single-seed version of this figure reported a near-oracle GFlowNet value; applying the multi-seed standard of §6(v) to our own figure corrected it to the value shown, a concrete instance of why that standard is mandatory. The decisive comparison against the stronger *iterative* heat-bath CI is Fig. 7.

quantum-sample reward under noise, §7, and genuinely multireference systems), not on this noise-free, cheap-reward illustration, and certainly not against the strong *iterative* heat-bath CI of Fig. 7 in §5, where the classical selector wins at every FCI-verifiable scale.

4.7 A note on real-space neural wavefunctions

For completeness and to forestall a common conflation, we distinguish the methods above from the real-space neural-wavefunction programme (FermiNet, PauliNet, Psiformer, and the emerging foundation models Neural Pfaffians and Orbformer) [71–75]. These are variational Monte Carlo ansätze in continuous three-dimensional space, not determinant selectors in a fixed orbital basis; they are among the strongest classical methods for small strongly correlated molecules and thus appear in §5 as competitors. The discrete-orbital neural-network backflow of Liu and Clark reaches state-of-the-art molecular energies among them [76]; its optimized variant now matches heat-bath CI, ASCI, FCIQMC and DMRG and surpasses coupled cluster, making it the strongest single learned proposer to date [77] (a related neural ansatz reaches high accuracy with only a handful of Slater determinants [78]), and the programme is surveyed by [79]. But these methods do not propose

configurations for a diagonalization and are outside the loop this review concerns. The one bridge worth flagging is the discrete-basis autoregressive NQS (NAQS) [80], which *does* sample determinants proportional to $|\psi|^2$ and is therefore the closest classical relative, and the most natural benchmark, for any learned determinant proposer, GFlowNet-based or otherwise.

5. The critical assessment: does the quantum sampler, or its ML augmentation, beat classical selected CI?

Every method in §4 is ultimately in service of an SQD subspace, and every SQD subspace inherits a single, unavoidable question. Because the final step is an exact diagonalization in a determinant subspace, SQD *is* a selected configuration interaction whose determinants happen to be chosen by a quantum sampler. Its natural competitors are therefore the classical selected-CI heuristics — and those are neither weak nor slow. A review that omitted this comparison would be advocacy, not assessment; we make it the centre of the paper.

5.1 The classical baselines and the accuracy bar

The relevant classical methods fall into two groups. The direct competitors are the classical determinant selectors: CIPSI, with its Epstein–Nesbet perturbative selection and extrapolation to the FCI limit [2, 81, 82]; heat-bath CI (HCI) and its semistochastic form (SHCI), whose $|H_{ai}C_i|$ screening makes selection near-linear and which is generally the fastest route to sub-milli-hartree accuracy [3, 4]; and adaptive sampling CI (ASCI) [5]. The best-in-class solvers for strong correlation are DMRG [83, 84], phaseless AFQMC [85], full configuration-interaction quantum Monte Carlo [86], and, for weak-to-moderate correlation, coupled cluster. Each of these baselines has a canonical, openly available implementation against which any claim can be reproduced (Dice for semistochastic heat-bath CI, Quantum Package for CIPSI, `block2` for DMRG, `ipie` for AFQMC, and NECI for FCIQMC, all interoperable through PySCF), and we take it as part of the standard below that a benchmark name the baseline code it was run against. The accuracy these methods agree on sets the bar any quantum result must clear to matter, and that bar is now quantitative: in the blind benzene ground-state challenge (a (30e,108o) space across seventeen groups), DMRG, AFQMC, FCIQMC, and SHCI concur on the correlation energy to a root-mean-square deviation of 1.3 mHa [87], and the Simons hydrogen-chain benchmark reaches comparable cross-method concordance in the thermodynamic limit [88]. The same collaboration’s transition-metal capstone (twenty first-principles methods on 3d atoms, ions, and monoxides, with experiment-free reference values [89]) fixes the classical bar in exactly the strongly-correlated, 3d/iron–sulfur arena that SQD most wants to claim, and a curated open hierarchy of chemically decisive multireference systems (N_2 , FeS, [2Fe-2S], U_2) has since been assembled to standardize such comparisons [90]. The operative standard, then, is not “reaches chemical accuracy” but “reaches the ~ 1.3 mHa cross-method consensus at *lower cost* than these methods, or in a regime where they all degrade.” No SQD result has yet been measured against this bar and passed; indeed the largest systematic hardware benchmark to date (745 reactions on a superconducting processor) finds that SQD attains CCSD-level accuracy only after energy extrapolation, not directly [91].

5.2 The critique literature

Two peer-reviewed-track critiques frame the negative case. Reinholdt and co-workers, granting SQD an idealized noiseless sampler, show on N_2 and [2Fe-2S] (the very systems used to showcase the method) that its expansions are roughly an order of magnitude less compact than heat-bath CI (for [2Fe-2S], on the order of 5×10^4 determinants against heat-bath CI’s $\sim 4,300$ at matched accuracy), and that the coupon-collector cost to reach micro-hartree precision extrapolates to $\sim 10^{14}$ samples; their conclusion is that QSCI “falls behind more effective classical selected-CI heuristics” [14]. Gaberle and Jattana sharpen the claim for strongly correlated lattice models: the

number of configurations required for a fixed accuracy grows *exponentially with system size even under optimal, decreasing-probability inclusion*: even a *perfect* sampler fails, so the limitation is intrinsic wavefunction delocalization, not a sampling artefact [24]. This second result is the more fundamental, and to date it has no proposed remedy.

5.3 The 2026 dequantization

The critique became concrete in 2026. Belagali and co-workers gave a polynomial-time classical algorithm for the energy of any single-layer local unitary cluster Jastrow circuit (the exact ansatz SQD samples), exploiting the fact that such a layer is a free-fermionic (matchgate) orbital rotation dressed by a low-rank diagonal-Coulomb (Jastrow) factor: the number–number Jastrow interactions are not themselves Gaussian, but at single-layer depth an extended-matchgate / fermionic-linear-optics treatment renders the energy classically tractable. The mechanism is specific: the Jastrow factor is diagonal in the occupation-number basis and low-rank, so the reduced density matrices needed for the energy collapse to Pfaffian-style evaluations over the free-fermionic vacuum rather than incurring the exponential cost a generic non-Gaussian layer would — which is exactly why the single-layer result does not extend to deep, multi-layer circuits. They reproduced the flagship 77-qubit [4Fe-4S] experiment on a laptop in under a minute, finding, through classical circuit optimization, an energy *below* the hardware result [26]. The authors are careful that this is weak (energy) rather than strong (bitstring) simulation, leaving a formal gap; but as a statement about the specific published experiments it is decisive. Independently, a classical iterative refinement seeded from random determinants (TrimCI) reached the [4Fe-4S] ground state with on the order of 10^6 -fold fewer determinants than the quantum run [92], and the noise-and-recovery study of §3.2 showed random determinants plus classical recovery reproducing the accuracy [25]. Taken together, these results locate the accuracy not in the quantum sampler but in the classical recovery-and-diagonalization back end. The trend has since consolidated. A fermionic-linear-optical (extended-matchgate) simulation not only reproduces the LUCJ circuit classically but *improves* on the noisy-hardware SQD energy [93]; this is unsurprising given device noise, the substantive point being that no quantum resource was needed, not that a new accuracy record was set. And, most tellingly, a bespoke classical eigenwalk solver from the same IBM authors who reported the 49-qubit advantage reportedly solves a subset of the instances underlying that claim classically [94], so the advantage question is now partly self-refereed within a single group. In parallel, classical coupled-cluster and DMRG reached chemical accuracy on the FeMo-cofactor model long held up as the canonical quantum target [95], and transformer neural-network backflow surpassed DMRG on [2Fe-2S], the very iron–sulfur system SQD showcases [96].

5.4 The pro-quantum case, weighed on its own terms

The defence is real but narrow. Hafid and co-workers prove that UCJ circuits can encode arbitrary instantaneous-quantum-polynomial (IQP) computations, so *some* such circuits are classically hard to sample unless the polynomial hierarchy collapses [27] — a worst-case statement that does not describe the few-layer, chemically-structured LUCJ circuits actually run, which §5.3 shows are easy. Weaving and co-workers report time-evolved QSCI subspaces more than two hundred times more compact than a naive selected-CI criterion and “comparable to HCI,” but explicitly decline to claim a quantum advantage [46]. The strongest experimental claim is Kirby and co-workers’ 49-qubit sample-based Krylov result, reported to outperform classical sparse-ground-state solvers [28] — but on *sparse-ground-state local Hamiltonians*, not general chemistry, a scope that must be stated precisely and whose classical rebuttals should be watched. The one genuinely open crack is depth: the dequantization proofs are for single-layer LUCJ, and whether deep, multi-layer LUCJ evades classical simulation is unresolved. It is the only place a future chemistry advantage could currently hide.

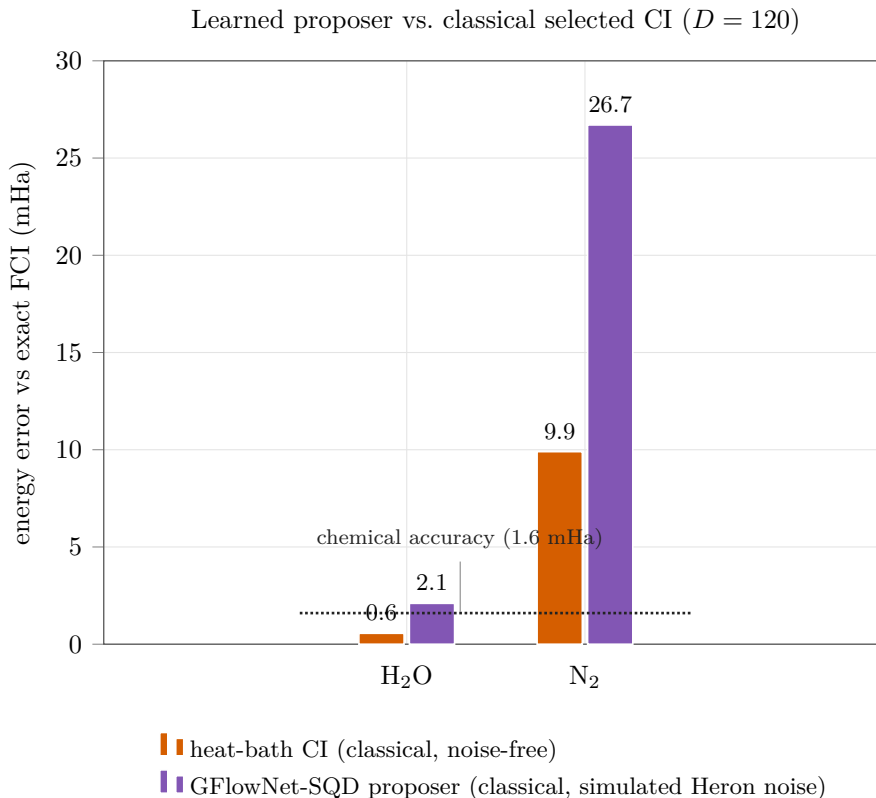


Figure 7. Classical selected CI versus a learned proposer at FCI-verifiable scale ($D = 120$; dotted line: chemical accuracy, 1.6 mHa). Iterative heat-bath CI, grown from the Hartree–Fock reference by Epstein–Nesbet perturbative selection (never seeded from the exact wavefunction), reaches 0.6 mHa on H₂O and 9.9 mHa on stretched N₂, below the fused GFlowNet-SQD subspace built under simulated Heron noise (2.1 and 26.7 mHa). This pits a noise-free classical selector against a noise-affected learned proposer, so it illustrates the gap at realistic conditions rather than isolating the proposer from the noise; the controlled, noise-matched test is the study §6 calls for. It is an illustration on two FCI-verifiable systems, not the evidentiary basis of the §5.5 verdict, which rests on the independent literature.

5.5 Verdict

The current state of the field, as of mid-2026, is that there is *no reproducible, same-active-space demonstration that the SQD/QSCI quantum sampler beats SHCI, CIPSI, DMRG, or AFQMC (or even random determinants passed through classical recovery) on molecular electronic structure*, and that the flagship sampling circuits have been classically reproduced at the energy (weak-simulation) level. This is not a claim that SQD is useless: it is a genuinely useful, noise-tolerant, well-tooled workflow that has reached chemically relevant scale, and its configuration-recovery idea is a real contribution. It is a claim that the burden of proof for *advantage* rests squarely on the quantum side, that this burden has not been met, and, the point of §6 and §7, that meeting it, or ruling it out, requires a standard of benchmarking the field has not yet adopted. This verdict rests on the independent critique and dequantization literature above (not on any single computation of ours) and now coincides with independent macro-assessments of quantum advantage in electronic structure [97–99]. Fig. 7 merely illustrates it on two FCI-verifiable systems, where iterative heat-bath CI — grown from the Hartree–Fock reference by Epstein–Nesbet perturbative selection, each iteration re-scored against the current (increasingly correlated) wavefunction and never seeded from the exact solution — reaches 0.6 mHa on H₂O and 9.9 mHa on stretched N₂, below both the noisy fused GFlowNet-SQD subspace (Fig. 7) and the noise-free cheap-reward GFlowNet subspace (Fig. 6).

6. A standard for honest benchmarking

The pattern that recurs across §5 is not that quantum methods were shown to fail, but that they were often not measured against the baselines that would settle the question. We therefore propose, and argue the field should adopt, a benchmarking standard with ten elements; we state it as a checklist because its value is in being applied uniformly. Systematic hardware benchmarking of SQD has recently begun [91]; our emphasis differs in insisting on exact-FCI ground truth and an explicit dequantization baseline, and the checklist is exactly the protocol any defensible study of a learned proposer — ours or others’ — would follow.

(i) Exact ground truth at verifiable scale. Claims about which determinants matter, and how compact a subspace is, are only meaningful where the exact answer is known. The controlled experiment should live in active spaces small enough for exact FCI (up to roughly a dozen orbitals) — large enough for the coupon-collector effect to appear, small enough that every energy is an error against truth, not against another approximation. This is a deliberate inversion of the “largest active space” instinct: the point is not scale but falsifiability.

(ii) The strong classical baseline, not a straw man. The comparison must be against iterative heat-bath / semistochastic CI bootstrapped from the correlated wavefunction — and, following §5.3, against the two baselines the field has repeatedly found decisive: random determinants passed through the classical recovery loop, and the classical simulation of the sampling circuit itself. A method that beats uniform sampling but not SHCI has demonstrated nothing about quantum value.

(iii) A dequantization baseline. Given [26], any hardware claim should report whether the sampling circuit is classically simulable at the depth used, and if so, the energy that classical simulation attains. This is now as basic a control as reporting the classical FCI reference, and it has four independent routes the strongest studies report against wherever they apply: fermionic-linear-optical (matchgate) simulation [26, 93], sparse guided eigenwalks [94], Pauli propagation [100], and direct tensor-network (matrix-product-state) contraction of the circuit — the last two being the depth-truncable routes that most directly probe the multi-layer-LUCJ crack of §5.4.

(iv) Full cost accounting in the currency that is scarce. The scarce resource is quantum measurement (shots = QPU time), and the honest axis is energy error versus shots, or versus subspace dimension at matched classical diagonalization cost — not energy error versus wall-clock on incomparable hardware. Reporting the shot budget honestly also surfaces the early saturation of §3.3.

(v) Statistical error bars. Sampling is stochastic and neural proposers are seed-dependent; a single run is an anecdote. Multi-seed means and standard deviations, and a significance statement on any claimed separation, should be mandatory. In our own controlled work we found that a single-run “win” of a generative proposer over a classical control at moderate ($1\times$ Heron) noise evaporated to a statistical tie (gap 0.3 ± 2.4 mHa over five seeds) once error bars were computed; the same five-seed analysis shows a separation emerging only at higher noise (Fig. 10), which is why we report that crossover as regime-dependent rather than as a win — a cautionary instance of exactly this point.

(vi) Backend-calibrated, asymmetric noise. Because SQD’s noise sensitivity runs through particle-number-violating readout errors (§3.2), a credible noise model must be backend-calibrated and must include the T_1 -driven asymmetry of readout, which symmetric bit-flip models understate; and the analysis should include an ablation isolating that asymmetry. A readout-only model will, in turn, be criticized where gate errors dominate, so the strongest studies will move toward full noisy-circuit simulation or a confirmatory hardware point.

(vii) Convergence of the reduction to the model problem. Every SQD number is a claim about a chosen active space in a chosen basis; the benchmark should report active-space, basis-set, and orbital-choice (natural- versus canonical-orbital) convergence, so that an apparent advantage is not an artefact of a fortunate truncation.

(viii) Energy differences and properties, not only total energies. Chemistry is decided by

Where a quantum or generative advantage can, and cannot, live

The structure that makes SQD classically *verifiable* is the structure that makes it classically *constructible*. Advantage survives only where that equivalence breaks.

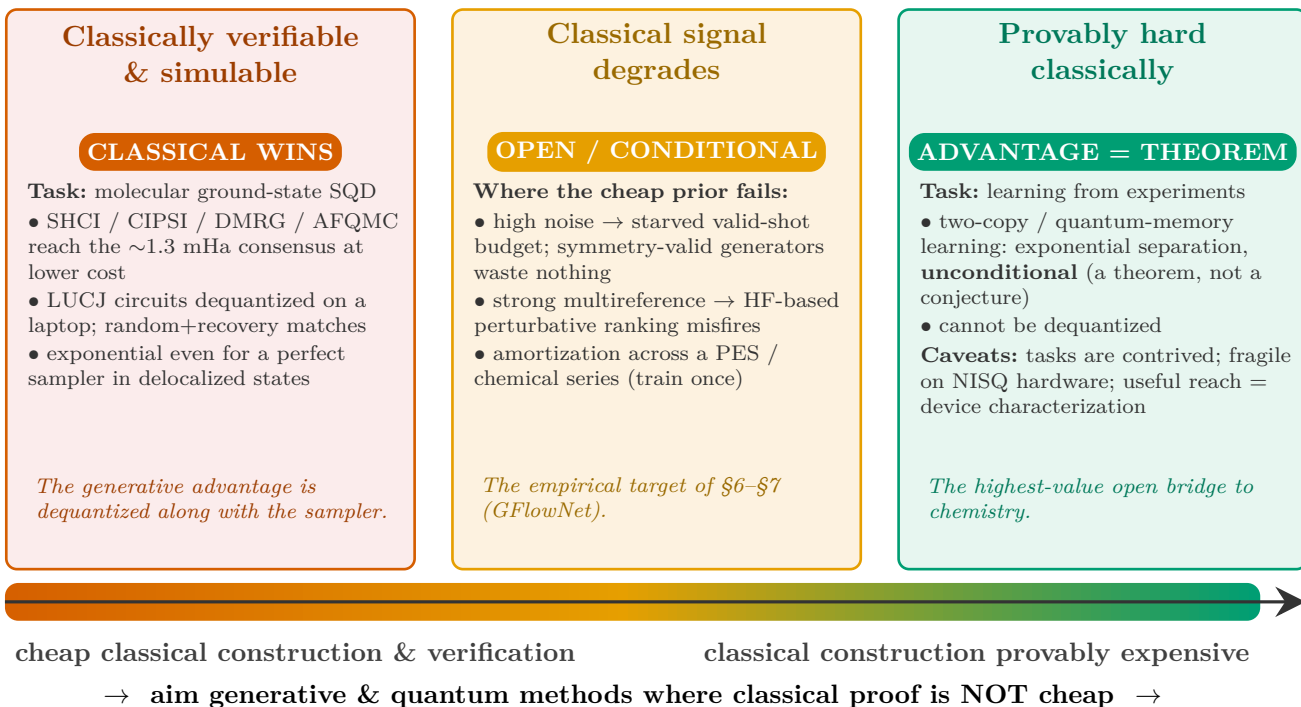


Figure 8. The advantage landscape of §5–§7. The structure that makes SQD classically *verifiable* is the structure that makes it classically *constructible*; advantage survives only where that equivalence breaks. Colour-blind-safe (Okabe–Ito) palette; zones are also distinguished by header text and verdict chips, not colour alone.

relative quantities — barriers, spin gaps, dissociation energies — and a method can reach a good total energy while misranking the differences that matter; a benchmark should report at least one chemically decisive energy difference, not a single total energy.

(ix) More than one geometry. A single-point win can be a coincidence of the reference; several geometries along a dissociation or reaction coordinate test whether the advantage is systematic, and are the natural setting for the amortization question of §7.

(x) Pre-registered, commensurate total-resource accounting. Beyond the shot axis of (iv), the true total cost is quantum and classical together — QPU-seconds reported alongside the classical node-hours of the diagonalization and recovery, in the commensurate-resource spirit of [91] — and the seed count and significance threshold of (v) should be fixed before the runs, not chosen after.

We contend that a study meeting all ten is publishable and valuable *whatever its result* — a rigorous negative result that maps where a generative or quantum proposer does and does not help is, in a field this contested and this prone to under-benchmarked demonstrations, more useful than another hopeful positive that a strong baseline would deflate.

7. Outlook: where generative and quantum methods retain a defensible advantage

The verdict of §5 is easy to read as a counsel of despair; we read it instead as a map. The reason classical selected CI wins the SQD contest, on the systems studied, is structural: SQD’s task is *classically verifiable*, and, empirically on molecular systems, the same structure that lets one check a subspace (a projected Hamiltonian one can diagonalize) also makes a cheap classical *importance*

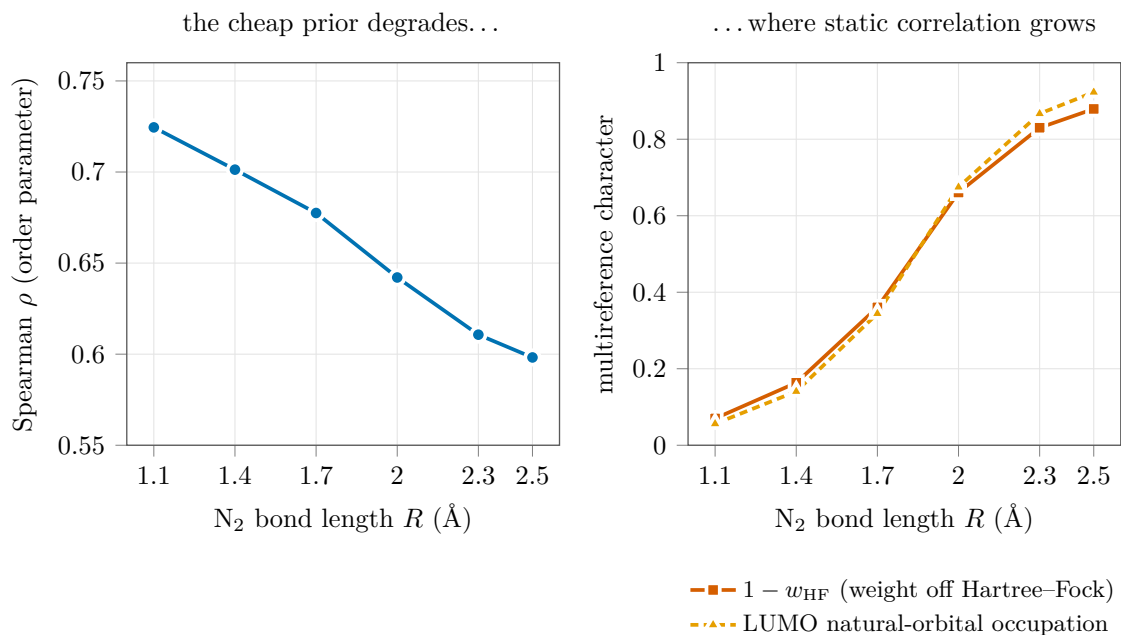


Figure 9. The order parameter, measured on N_2 across geometries. (a) The Spearman rank correlation between the cheap Epstein–Nesbet reward and the exact FCI $|c|^2$ (over all 792 α -strings) falls monotonically as the triple bond stretches, from $\rho=0.72$ near equilibrium to 0.60 at $R=2.5$ Å. (b) Over the same interval the wavefunction becomes strongly multireference: the weight carried by determinants *other* than Hartree–Fock rises from 0.07 to 0.88, and the frontier natural-orbital occupations open from the closed-shell 2/0 toward the dissociation limit 1/1 (LUMO occupation 0.06 $\rightarrow$ 0.92). The cheap classical importance signal that makes an SQD subspace *verifiable* degrades exactly where static correlation makes it hard to *construct* — so the single scalar ρ predicts, before any hardware is run, where a learned or quantum proposer has room to help.

heuristic available with which to construct it; the dequantization of §5.3 is the extreme statement of the same fact, that even the sampler is classically reproducible. (This is an observation about the available classical signal, not a general implication of verifiability: verifiability alone need not imply cheap constructibility, as the P-versus-NP gap makes plain.) There is little quantum edge to be had where such a cheap classical construction happens to exist.

This is not a wall but a compass: it says to aim generative and quantum methods at the regimes where a strong classical baseline provably *cannot* be cheap. This converts the compass from a metaphor into a falsifiable prediction the standard of §6 can test: if the locus of advantage is exactly where the cheap importance signal misranks the truly important determinants, then any genuine generative or quantum advantage on molecular electronic structure should appear first, and only, where the *rank correlation between the cheap prior (e.g. the Epstein–Nesbet coefficient) and the exact CI weights collapses*. That rank correlation is directly measurable at FCI-verifiable scale, and our own minimal computational experiment both confirms it is imperfect and shows it behaving exactly as an order parameter should. Over the 792 α -strings of the $N_2(10e,12o)$ active space, the Spearman correlation between the cheap Epstein–Nesbet reward and the exact $|c|^2$ is $\rho \approx 0.64$ at $R = 2.0$ Å; more tellingly, it *degrades monotonically as the wavefunction becomes more multireference*, falling from $\rho=0.72$ near equilibrium ($R = 1.1$ Å: Hartree–Fock weight 0.93, frontier natural-orbital occupations 1.95/0.06) through 0.64 at $R = 2.0$ Å (HF weight 0.34, occupations 1.32/0.68) to 0.60 at $R = 2.5$ Å (HF weight 0.12, occupations 1.08/0.92, the sum of fractional occupancy rising from 0.29 to 5.10). This is the order parameter measured (Fig. 9): the single scalar collapses precisely where static correlation grows, i.e. where a single-reference prior must misrank the determinants that matter. We propose it as the coordinate for where to look — a scalar computable, system by system, before any hardware is run. A method that helps where the prior already ranks well would falsify the picture; a method that helps only as this correlation collapses would confirm it. Three such regimes are visible.

Noise-limited recovery and genuine multireference character. Within SQD itself, the classical selectors win because a cheap classical importance signal (a Hartree–Fock-referenced perturbative estimate) is available and accurate. Where that signal degrades (in the high-noise regime, where the valid-shot budget is starved and a generator that emits only symmetry-valid configurations by construction wastes nothing, and in genuinely multireference systems, where a single-reference perturbative prior misranks the important determinants), a learned proposer that fuses the quantum-sample statistics with a cheap classical prior has room the *perturbatively-primed* classical selector does not. (Here the fair baseline is not only iterative SHCI but the classical learned rankers of §4.4, which already contest, and on Cr₂ reportedly beat, SHCI in exactly this multireference regime with no quantum resource; the live question is therefore the *incremental* value of the quantum samples fused into such a learned proposer, not of learning over perturbation theory per se.) Crucially, the fair classical control here is itself the noise-robust recovery loop of §5.3 (S-CORE plus the cheap prior that already reproduces the accuracy), so any generative gain is measured *beyond* that classical recovery. Consistent with §6(v), that gain is a statistical tie at low-to-moderate noise and reaches significance (a paired $t_4=8.7$, $p\approx 0.001$ over five seeds, i.e. a Gaussian-equivalent $\sim 3.3\sigma$) only above $\sim 2\times$ Heron noise (Fig. 10); even there it is a conditional effect, confined to the multireference system and at energies far from chemical accuracy, not a reversal of the verdict. Its *multireference* half — that the advantage should appear *only* where a single-reference prior misranks the important determinants — we can now partially test, because our two FCI-verifiable systems differ in precisely that variable. Repeating the noise sweep on *single-reference* H₂O (Hartree–Fock weight 0.96, versus 0.34 for N₂) yields *no* crossover: the classical control wins or ties at every noise scale and is significantly *better* at $2\times$ (-2.9 ± 1.0 mHa, paired $t_4=-6.4$), despite suffering the same valid-shot starvation (53% lost at $3\times$; Fig. 10). The generative advantage therefore tracks multireference character, not noise alone — exactly as the order parameter predicts. This is still two molecules at energies above chemical accuracy, not a systematic study; extending it along the full order-parameter axis (e.g. on Cr₂ or a metal dimer) is the controlled, exact-ground-truth study §6 calls for, and the natural first empirical target for the GFlowNet route of §4.6. We emphasize that the expectation, from the trajectory of §5, is a *conditional* advantage (a regime map, not a universal win), and that a paper reporting such a map, negative regions included, is the valuable contribution. Fig. 10 is that map in miniature, on two FCI-verifiable systems: for multireference N₂ the classical-minus-generative gap is *negative* at zero noise (the classical prior alone suffices) and turns positive as the device-noise scale grows and the valid-shot budget starves, saturating into a plateau once that budget is exhausted; for single-reference H₂O it stays negative throughout. The mechanism, sharpened by the H₂O control, is a *conjunction*: the constraint-respecting proposer emits only symmetry-valid configurations and so wastes nothing as the valid-shot fraction collapses (Fig. 4), but this buys an advantage *only* where the cheap prior is also intrinsically imperfect. Where the prior is near-exact (single-reference H₂O), the classical control wins even under heavy shot starvation; valid-shot starvation is necessary but not sufficient, and multireference character is what completes the condition.

Amortization across chemical space. A learned proposer offers something a per-instance classical selector does not: amortization. A model conditioned on molecular geometry and trained once can, in principle, generate compact subspaces across an entire dissociation curve or chemical series without re-running an expensive selection at each point — the value proposition of the foundation-wavefunction programme [75], transposed to the determinant-selection setting. Here the metric is not accuracy at one geometry but total cost across many, and it is a genuinely open, and genuinely ML-flavoured, question whether amortization beats re-running SHCI per point.

Learning from experiments: where the advantage is a theorem. The deepest reason for optimism lies just outside the SQD loop, and it inverts the logic of §5 exactly. The one place in this landscape with an *unconditional, information-theoretic* quantum advantage is the quantum-data-native task of learning properties of an unknown quantum state or process: a learner with quantum memory that measures two coherent copies jointly needs exponentially fewer experiments than any single-copy strategy, a separation proved without complexity assumptions [101] and demonstrated on superconducting and photonic hardware [102, 103], and surveyed among the field’s few unconditional

The noise crossover: generative advantage is regime-dependent, emerging only under noise

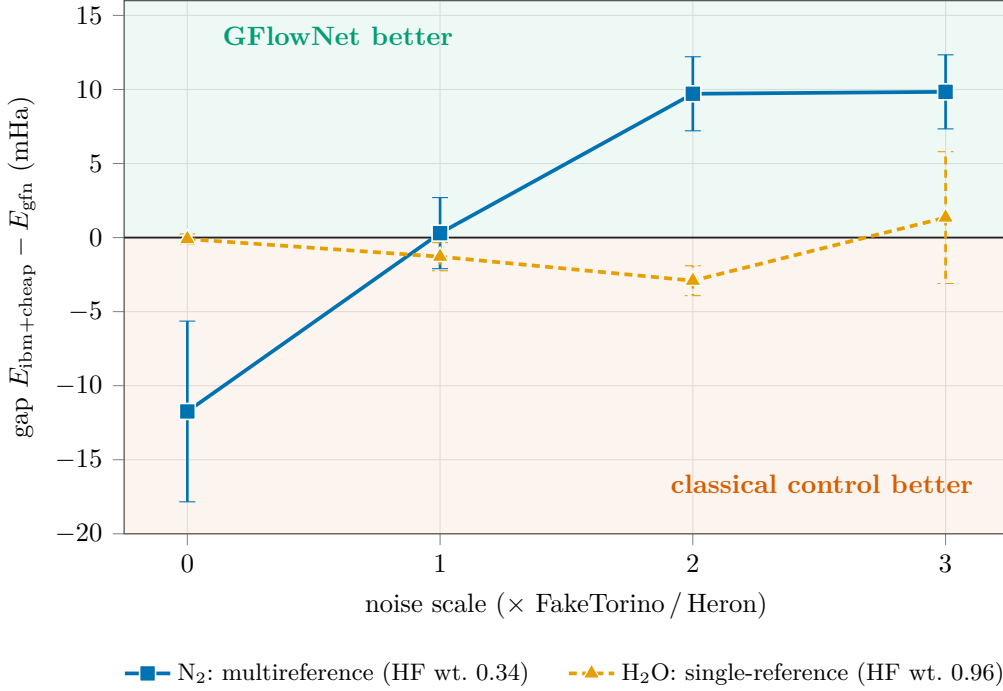


Figure 10. The noise crossover is specific to the multireference regime (two FCI-verifiable systems). Energy-error gap between the fair classical control (S-CORE recovery + cheap-prior fill) and the fused generative proposer, vs *simulated* device-noise scale (FakeTorino / Heron r1), as mean $\pm$ s.d. over five seeds. *Strongly multireference* stretched N₂ (blue; Hartree–Fock weight 0.34) shows the crossover: the classical control is better or tied at low noise (-11.7 ± 6.1 at $0\times$, 0.3 ± 2.4 mHa at $1\times$) and the constraint-respecting generative proposer pulls ahead only at higher noise (9.7 ± 2.5 and 9.8 ± 2.5 mHa at $2\times$ and $3\times$; paired $t_4=8.7$, $p\approx 0.001$, Gaussian-equivalent $\sim 3.3\sigma$ — not the naïve $9.7/\text{s.d.}$ effect size). *Single-reference* H₂O (orange; HF weight 0.96) shows *no* crossover: the classical control wins or ties throughout, and is in fact significantly better at $2\times$ (-2.9 ± 1.0 mHa, paired $t_4=-6.4$), even though H₂O suffers the same valid-shot starvation (53% of shots lost at $3\times$, versus 51% for N₂). The contrast isolates the mechanism: valid-shot starvation alone does not produce a generative advantage; the crossover requires *both* noise *and* an intrinsically imperfect cheap prior, i.e. multireference character (§7). A conditional, regime-dependent effect at energies above chemical accuracy — not a universal win.

advantages by [104]. Unlike SQD, this advantage cannot be dequantized, because its classical lower bound is a theorem, not a conjecture about circuit structure. This suggests a reorganizing principle for the enterprise as a whole: candidate quantum advantages should be sorted not by their promised magnitude but by the *type* of their classical lower bound. Those resting on a complexity-theoretic conjecture (the presumed hardness of sampling a particular circuit family) are perpetually one clever classical algorithm away from collapse, as SQD’s own two-year arc from flagship experiment to laptop reproduction (§5.3) attests; those resting on an unconditional, information-theoretic separation are not, because no future algorithm can close a gap that is already a theorem. A field repeatedly surprised by dequantization would do well to spend its scarce hardware where the lower bound is a theorem, and to treat conjecture-backed advantages as demonstrations of scale, not of separation, until proven otherwise. Its caveats are equally sharp and belong in any forward-looking account: the demonstrated-advantage *tasks* (predicting Pauli expectations, purity, and random-channel characteristic functions) are information-theoretically natural but scientifically contrived; the advantage is fragile on superconducting hardware, degrading under the depolarizing noise of a heavy-hex device [105] to the reduced, mitigation-propped separation IBM has actually measured [106]; and the one *useful* application delivered with rigorous backing is the characterization of quantum devices themselves. The bridge from this provable-but-contrived frontier to genuinely

useful chemistry is, we argue, the highest-value open problem adjacent to the subfield, and the natural destination for the generative machinery reviewed here. That bridge might be built by learning a device’s effective error model (as machine-learning quantum error mitigation already does at utility scale [107]) to improve an SQD estimator, or by learning the properties of a molecular state prepared on hardware.

8. Conclusions and open problems

Machine learning entered sample-based quantum diagonalization for a sound reason (the configuration-recovery step is a selection-and-denoising problem that a learned model should do better than an occupancy-weighted heuristic), and in two years has produced a rich ecosystem of restricted Boltzmann machines, neural-quantum-state proposers, classifiers, ranking models, and generative circuit designers. This review has organized that ecosystem, mapped its methods and open-source tooling, and, most importantly, insisted on holding it to the question it exists to answer. That verdict, stated in §5.5, is at present a negative one: no verifiable-scale demonstration that the quantum sampler or its machine-learning augmentation beats classical selected configuration interaction on molecular electronic structure, and flagship sampling circuits now classically reproducible at the energy (weak-simulation) level. We have argued that it is a compass rather than a counsel of futility: it points generative methods toward the noise-limited and multireference regimes where the cheap classical signal fails — a direction for which we now provide first controlled, FCI-exact evidence, since the prior’s rank correlation with the truth collapses with multireference character and the generative noise-crossover appears only in the multireference system and not the single-reference control (§7) — toward amortization across chemical space, and, as an adjacent horizon, toward the quantum-data-native learning tasks where the advantage is provable but not yet connected to useful chemistry.

Four open problems structure the road ahead. First, the *intrinsic-delocalization* barrier of [24] (that some strongly correlated ground states require exponentially many configurations even under a perfect sampler) has no proposed remedy and may be fundamental. Second, the *depth* question: whether deep, multi-layer LUCJ circuits evade the classical simulability that [26] proved for the single-layer case is the one place a chemistry advantage could still live, and settling it is urgent. We flag a structural worry worth confronting rather than postponing. The VQE programme collapsed when it emerged that the conditions letting a circuit avoid barren plateaus tend also to render it classically simulable [9]; SQD may face an isomorphic dichotomy in a different dialect, since the shallow, structured, low-entanglement circuits whose samples are noise-tolerant and amenable to configuration recovery are precisely those most exposed to matchgate-style dequantization [26], while the depth that would defeat classical simulation may also be the depth at which readout noise and the coupon-collector tail (§3.1) overwhelm the loop. Whether a usable corridor exists between these two failure modes is, we conjecture, the question on which a near-term *chemistry* advantage for SQD ultimately rests, and it is one the field has the tools to settle. Third, the *useful-target* gap in learning-from-experiments (bridging provable-but-contrived tasks to science anyone needs) is the highest-value problem adjacent to the field. Fourth, and most immediately actionable, the *GFlowNet gap*: the one generative paradigm built for diverse, reward-proportional sampling over exponential discrete spaces has not been applied to determinant selection, and whether it helps, measured (per §6) against exact ground truth and strong classical baselines, is a clean, falsifiable, and currently unclaimed question. We have tried to write the review such that its most useful legacy would be not a catalogue but a standard: that the next claim in this field, positive or negative, arrives with exact ground truth, strong and dequantization baselines, full cost accounting, and error bars, so that the field can, at last, know what it knows.

Limitations of this review. This survey is a snapshot of a subfield that is expanding on a timescale of weeks; a substantial fraction of the primary literature we cite is preprint and not yet peer-reviewed. We have verified the authors, venues, and arXiv/DOI identifiers of the works most central to the verdict, but preprint metadata can still shift, and readers should confirm against the published

versions of record. Our coverage extends to mid-2026 and will inevitably miss work that appears between writing and publication, including, very possibly, the first application of a generative-flow network to determinant selection whose absence we here record. We have tried to make the review robust to that churn by organizing it around durable questions (compactness, verifiability, the locus of advantage) and a benchmarking standard, rather than around any single result. Finally, a positional disclosure: the author is actively developing the generative-flow-network proposer whose absence this review documents, and the standard of §6 is the protocol that study is intended to meet. The GFlowNet gap is therefore presented as an open, falsifiable question (one that forthcoming study may well answer in the negative) rather than a solved one, and the §5 verdict is rested on the independent literature rather than on the author’s own illustrative figures.

Data and code availability

Every quantitative result we compute ourselves in this review — as distinct from the numbers we quote from the cited literature — is reproduced from executable code in the companion computational notebook `GFlowNet_SQD_calculations.ipynb`, which runs in Google Colab or any Python ≥ 3.10 with `pyscf`, `torch`, `scipy`, and `matplotlib`. The notebook and calculation scripts regenerate, from first principles and with fixed random seeds, the exact FCI reference of the systems in Fig. 2, the coupon-collector statistics of §3, the S-CORE recovery of §3.2, the Epstein–Nesbet reward and GFlowNet compactness study of §4, the natural-orbital occupations and order-parameter sweep (Fig. 9) together with the two-system noise crossover (Fig. 10) of §7, and the heat-bath-CI decisive test of §5. The complete code — the notebook, the calculation scripts behind every reported number, one script per figure, the generated figures, and the Docker build environments — is openly available (MIT license) at <https://github.com/nicolasbonilla/ml-for-sqd-review>, with a one-click Google Colab entry point and every notebook cell and figure rendered inline in the browser. Fig. 1 was produced as native vector graphics; Fig. 2 combines vector molecular-orbital diagrams with high-resolution (≥ 300 dpi at print size) ray-traced (PyMOL) isosurface renders, and all figure sources are available with the code. No proprietary data or hardware access is required to reproduce any figure or table; all “real-noise” results use local, backend-calibrated noise models (FakeTorino, IBM Heron r1).

Author contributions

N.B.V. conceived the review, performed the literature synthesis, designed and ran all computational experiments, produced the figures, and wrote the manuscript.

Competing interests

The author declares no competing financial interests. As a non-financial interest, the author is developing a generative-flow-network determinant proposer for SQD of the kind this review identifies as a methodological gap (§4.6, §8). The review documents that gap from the published literature and rests its central verdict on independent results, not on the author’s own illustrative computations; this disclosure is made in the interest of full transparency.

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